Saahil Doshi

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Education

Oconee County High School Expected Graduation: May 2027 GPA: 103.89

Relevant Coursework: AP Physics C, AP Calculus AB/BC, AP Physics 1, Foundations of Engineering

Research Experience

University of Georgia Research Internship Summer 2024; Faculty Mentor: Dr. Ramana Pidaparti

- Conducted an 8-week research project comparing bio-inspired, generative AI-designed, and control airfoils using XFOIL/NeuralFOIL simulation, CAD modeling, and wind tunnel validation.
- Built Python-automated simulation workflows and experimental test rigs for wind tunnel testing.
- Through controlled experimentation, achieved peak lift-to-drag ratio of 62 and maximum lift of 6.76 N.
- Earned a Regional Science and Engineering Fair Gold Medal and advanced as a Georgia state finalist.

Engineering Projects

NAR Level 1 Certification and Modular Research Rocket 2025–Present **Project Lead**

- Designing a high-power modular research rocket to carry 132 germinated beans and study high-G ascent effects on early biological development while working toward NAR Level 1 certification.
- Developed a avionics bay and cross-compatible airframe standard with interchangeable nose cones, payload sections, and fin cans for rapid A/B testing and published a paper in the AIAA Region II Conference.

NASA Student Launch Initiative 2025 **Payload Lead**

- Designed payload systems for an 8-foot, 4-inch diameter launch vehicle, integrating avionics, structural components, and recovery interfaces, which reached approximately 4,300 ft apogee.
- Collaboratively authored 500+ pages of NASA-reviewed documentation and validated OpenRocket predictions using dual altimeters and GPS flight data.

American Rocketry Challenge (ARC) 2023–Present **Captain**

- Designed and flight-tested a rocket constrained to a target apogee, 43–46 s flight time, and sub-650 g.
- Achieved national qualification as the sole team from Georgia and placed 24th out of 1,000 teams.

Leadership, Work, and Technical Engagement

Civil Air Patrol, U.S. Air Force Auxiliary 2022–Present **Cadet Commander / Cadet Colonel**

- Lead weekly aerospace education and leadership sessions focused on rocketry, dynamics, and cadet development; selected for Region Cadet Leadership School and recognized as sole Distinguished Graduate.

Mathnasium 2026–Present **Math Instructor**, 12 hrs/week

- Teach math concepts to elementary through high school students, adapting explanations to individual learning styles and reinforcing problem-solving confidence.

Governor's Honors Program (Mechanical & Aerospace Engineering) Summer 2025

- Selected for Georgia's competitive residential research program; collaborated on an ionic propulsion aircraft project involving experimental testing, CAD integration, and iterative design under constraints.

Technical Skills

Python, LaTeX, JavaScript, Java | Fusion, AutoCAD, SolidWorks, OnShape | XFOIL & NeuralFOIL, OpenRocket, Finite Element Analysis, Computational Fluid Dynamics | MATLAB | Ansys

Activities

Young Dawgs Internship | Technology Student Association | FIRST Robotics | Varsity Tennis (Team Captain) | Student Council (Vice President) | National Honor Society | Beta Club | Spanish Club

Honors and Awards

AIAA Regional Conference 1st Place | President's Award for Academic Excellence | Regional Science and Engineering Fair Gold Medalist | 24th Place Nationally – American Rocketry Challenge | College Board AP Scholar with Distinction | Region Cadet Leadership School Honor Seminar & Distinguished Graduate